

The Ability of Critical Care Physicians to Identify Patient-Ventilator Asynchrony Using Waveform Analysis: A National Survey

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BACKGROUND: Improved patient-ventilator asynchrony (PVA) identification using waveform analysis by critical care physicians (CCPs) may improve patient outcomes. This study aimed to assess the ability of CCPs to identify different types of PVAs using waveform analysis as well as factors related to this ability. **METHODS:** We surveyed 12 university-affiliated medical ICUs (MICUs) in Tunisia. CCPs practicing in these MICUs were asked to visually identify 4 clinical cases, each corresponding to a different PVA. We collected the following characteristics regarding CCPs: scientific grade, years of experience, prior training in mechanical ventilation, prior exposure to waveform analysis, and the characteristics of the MICUs in which they practice. Respondents were categorized into 2 groups based on their ability to correctly identify PVAs (defined as the correct identification of at least 3 of the 4 PVA cases). Univariate analysis was performed to identify factors related to the correct identification of PVA. **RESULTS:** Among 136 included CCPs, 72 (52.9%) responded to the present survey. The respondents comprised 59 (81.9%) residents, and 13 (18.1%) senior physicians. Further, 50 (69.4%) respondents had attended prior training in mechanical ventilation. Moreover, 21 (29.2%) of the respondents could correctly identify PVAs. Double-triggering was the most frequently identified PVA type, 43 (59.7%), followed by auto-triggering, 36 (50%); premature cycling, 28 (38.9%); and ineffective efforts, 25 (34.7%). Univariate analysis indicated that senior physicians had a better ability to correctly identify PVAs than residents (7 [53.8%] vs 14 [23.7%], $P = .044$). **CONCLUSIONS:** The present study revealed a significant deficiency in the accurate visual identification of PVAs among CCPs in the MICUs. When compared to residents, senior physicians exhibited a notably superior aptitude for correctly recognizing PVAs. *Key words:* patient-ventilator asynchrony; waveform analysis; respiratory support; mechanical ventilation; identification accuracy; education and training; critical care physicians. [Respir Care 2024;69(2):176–183. © 2024 Daedalus Enterprises]

Introduction

Patient-ventilator asynchrony (PVA) is described as “a mismatch between neural and mechanical inspiratory times.”¹ A French study on 62 mechanically ventilated subjects showed that 24% had an asynchrony index > 10% of respiratory efforts.² It is a prevalent issue in critical care settings and can have important implications for patient outcomes. PVA can lead to patient discomfort, respiratory distress,³ prolonged mechanical ventilation,² a higher rate of unsuccessful weaning from mechanical ventilation,⁴ muscle injury, decreased sleep quality, and increased sedation or muscle relaxant demands, and even complications such as lung injury.^{5–11} Therefore, accurate identification and management of PVA are crucial for optimizing patient

care by improving patient comfort and, potentially, outcomes. Due to insufficient monitoring or medical professionals’ lack of experience, the incidence of PVA may be considerably understated. The frequency of PVA could be substantially undervalued due to insufficient monitoring or the lack of experience among physicians.¹²

Visual waveform analysis has emerged as a valuable bedside, noninvasive, and reliable method for assessing patient-ventilator interactions.^{12–16} By analyzing the graphical representation of pressure, flow, and volume waveforms generated by mechanical ventilators, clinicians can identify patterns indicative of various types of asynchrony. The number of asynchronous events detected by waveform analysis was demonstrated to be closely correlated with those detected through esophageal pressure measurement.^{2,17}

Understanding the factors that influence CCPs' ability to identify PVA through waveform analysis is crucial for several reasons. First, it can inform targeted educational interventions aimed at improving their skills in recognizing and addressing PVA. Second, identifying any associations between years of experience, prior training in mechanical ventilation, and proficiency can guide the development of training programs tailored to specific needs. Moreover, assessing the accuracy and reliability of CCPs' identifications can help determine the feasibility of integrating waveform analysis into routine clinical practice.

SEE THE RELATED EDITORIAL ON PAGE 275

Only a few studies have assessed the ability of health care professionals to correctly identify PVA. These studies revealed a low correct identification rate of PVA by CCPs and concluded that specific training in mechanical ventilation increased the ability of CCPs to identify PVA using visual waveform analysis.¹⁸ Therefore, this study aimed to assess the ability of CCPs to identify different types of PVA using visual waveform analysis. By analyzing a data set of anonymized waveform recordings from mechanically ventilated subjects, CCPs were evaluated for their accuracy and proficiency in identifying PVA patterns. Factors such as scientific grade, prior training in mechanical ventilation, or prior exposure to waveform analysis were examined to determine their influence on identification skills.

Methods

Study Design

This cross-sectional observational study included CCPs working at 12 university-affiliated MICUs in Tunisia. Ethical approval was obtained from the Research and Ethics Committee of the University Hospital Farhat Hached, Sousse. Informed consent was obtained from all participating CCPs. The present study followed the

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QUICK LOOK

Current knowledge

Waveform analysis is an established and reliable bedside method to assess patient-ventilator asynchrony (PVA). However, the proficiency of health care professionals in accurately identifying PVA varies widely between disciplines and years of experience.

What this paper contributes to our knowledge

This study highlights the limitations in correctly identifying PVAs among critical care physicians. Prior inconsistent training on ventilator graphics may have hindered the accurate identification of PVAs, indicating the need for more comprehensive and standardized training in this area and/or the implementation of ventilator built-in software to recognize and correct PVAs.

Consensus-Based Checklist for Reporting Survey Studies.¹⁹

Respondent Recruitment

All residents and physicians with active clinical involvement in managing mechanically ventilated patients in university-affiliated MICUs in Tunisia were considered eligible for participation in this study. Non-ICU residents who performed optional residency rotations in a MICU were not included. Demographic information, including scientific grade, years of experience, prior training in mechanical ventilation, and prior exposure to waveform analysis, was collected.

Data Collection

1. Waveform data set: A data set of anonymized waveform recordings from mechanically ventilated subjects was compiled. The data set included 4 clinical case scenarios of PVA patterns, namely ineffective efforts, auto-triggering, premature cycling, and double-triggering.
2. Survey: Respondents were invited to answer an online survey involving a 15-item questionnaire (Supplementary material, see related supplementary materials at <http://www.rjournal.com>).

Supplementary material related to this paper is available at <http://www.rjournal.com>.

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www.rcjournal.com). The questionnaire was sent by e-mail to respondents via SurveyMonkey. All responses were anonymous. Data collection began in July 2019 and ended in December 2019. Data collection did not exceed the 6-month resident rotation period in the same MICU. Multiple reminders were sent to ascertain the maximum number of respondents. Questions 1–11 involved demographic characteristics, including grade, years of experience, prior training in mechanical ventilation, and MICU characteristics. Questions 12–15 comprised 4 clinical case scenarios, which each contained a screenshot of a ventilator waveform showing different types of PVA (ineffective efforts, premature cycling, auto-triggering, and double-triggering). For each screenshot, CCPs were asked to choose the correct answer from 5 multiple choices.

Data Analysis

The 4 studied waveforms were chosen by a first expert (EE), then meticulously labeled by 2 blinded experts (MB, FT). The accuracy of CCPs' identification of different types of PVA was calculated by comparing their identification with a reference set at $\geq 3/4$ correct identifications. The accuracy rate is reported as a percentage. Respondents were divided into 2 subgroups, the proper identification group ($\geq 3/4$ correct identifications) and the improper identification group ($< 3/4$ correct identifications). Descriptive statistical analysis was conducted to examine factors related to identification skills, such as experience level, prior training in mechanical ventilation, and prior exposure to waveform analysis. Prior training in mechanical ventilation was defined as attending a mechanical ventilation course that contained at least one session of waveform analysis and PVAs. Prior exposure to waveform analysis was defined as a routine assessment of ventilator waveforms in their daily clinical practice.

Statistical Analysis

Statistical analyses were performed using SPSS software (IBM, Armonk, New York). Categorical data are presented as frequencies and percentages. Continuous data are expressed as medians and interquartile ranges.

Between-group comparisons of continuous and categorical variables were performed using the Mann-Whitney U test and chi-square test or Fisher exact test when appropriate, respectively. Statistical significance was set at $P < .05$.

Results

Respondents Characteristics

Respondents and respective MICUs characteristics are displayed in Table 1. Among 136 eligible physicians

Table 1. Characteristics of Respondents and ICUs Involved in the Survey Assessing the Ability of Critical Care Physicians to Identify Patient-Ventilator Asynchrony

Participant Characteristics	
Scientific grade	
Residents	59 (81.9)
Senior physicians	13 (18.1)
Prior training in mechanical ventilation	
Yes	50 (69.4)
No	21 (29.2)
Prior exposure to waveforms analysis	
Routine assessment	50 (69.4)
Occasional assessment	17 (23.6)
MICUs characteristics	
No. of MICU beds,	10 [8–19]
Number of residents	8 [5–8]
Number of senior physicians	5 [4–6]
Physician:bed ratio	1.33 [0.70–1.50]
Resident:bed ratio	0.69 [0.42–0.89]
Senior:bed ratio	0.59 [0.30–0.62]
Senior:resident ratio	0.81 [0.66–1.00]
Respondents practicing in MICUs admitting	66 (91.7)
< 500 patients/y	
Respondents practicing in MICUs admitting	6 (8.3)
≥ 500 patients/y	
Respondents practicing in MICUs admitting	54 (75)
< 200 ventilated patients/y	
Respondents practicing in MICUs admitting	18 (25)
≥ 200 ventilated patients/y	

Data are presented as *n* (%) or median [interquartile range].
MICUs = medical ICUs

working in the 12 MICUs of university-affiliated hospitals in Tunisia, 72 (52.9%) responded to the survey. Most respondents were residents 59 (81.9%); 50 (69.4%) had attended prior training in mechanical ventilation, and 50 (69.4%) reported prior exposure to waveform analysis. Most respondents were practicing in MICUs that admitted < 500 patients/y, 66 (91.7%), and < 200 ventilated patients/y, 54 (75%). The median number of MICU beds was 10 [8–19].

Identification Accuracy

Overall accuracy across all PVA types was 1.81 ± 1.26 of correct responses to the 4 clinical case scenarios of PVA patterns. There were 21 (29.2%) and 51 (70.8%) CCPs in the proper (≥ 3 correct responses) and improper (< 3 correct responses) identification groups, respectively. Specifically, 8 (11.1%), 13 (18.1%), 22 (30.6%), 15 (20.8%), and 14 (19.4%) CCPs correctly recognized 4, 3, 2, one, and zero PVAs, respectively. Double-triggering was the most frequently identified PVA type 43 (59.7%),

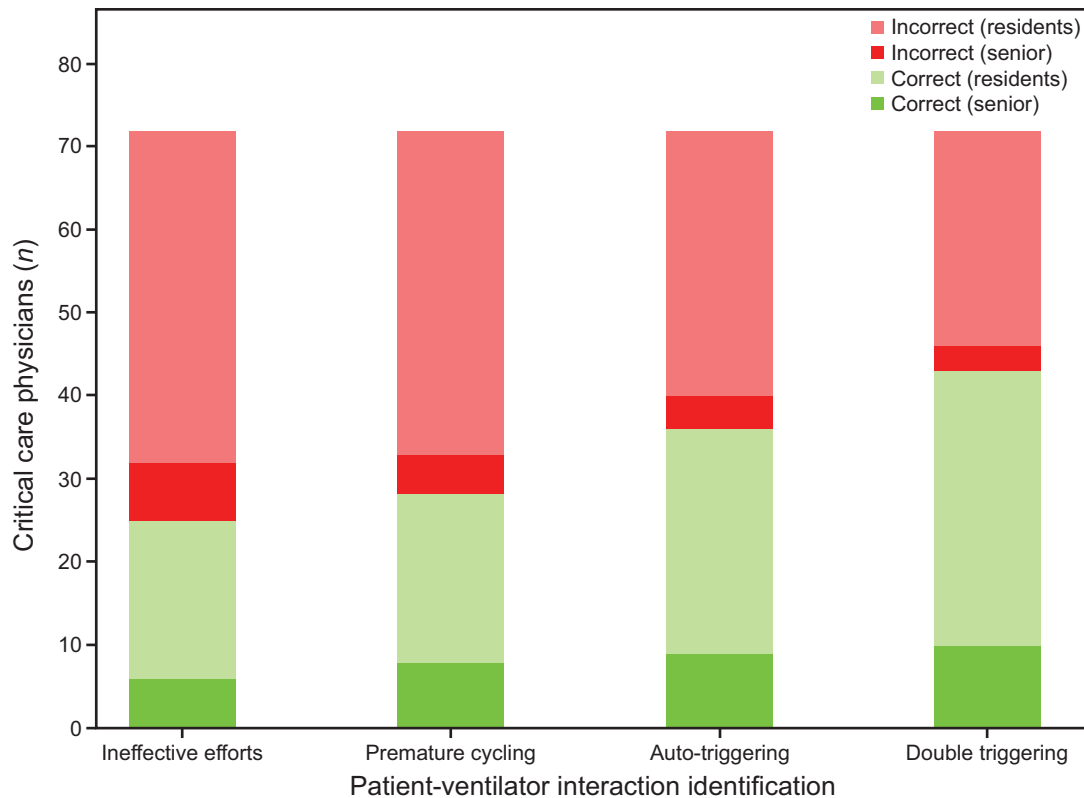


Fig. 1. Distribution of correct/incorrect identifications of patient-ventilator-asynchrony types among senior and residents critical care physicians. Double triggering was the most frequently identified patient-ventilator-asynchrony (PVA) type, followed by auto-triggering, premature cycling, and ineffective efforts. Seniors physicians demonstrated better identification skills among the respective PVA types.

followed by auto-triggering 36 (50%), premature cycling 28 (38.9%), and ineffective efforts 25 (34.7%) (Fig. 1).

Regarding waveform misidentification, “ineffective efforts” exhibited the highest frequency of misidentification; 21 of the 47 incorrect responses were confused with “flow asynchrony.” The remaining PVA misidentifications were as follows: premature cycling, 10/44 as flow asynchrony; auto-triggering, 10/36 as premature cycling; and double-triggering, 11/29 as no asynchrony.

Factors Related to Identification Ability

Table 2 summarizes the results of the univariate analysis between proper and improper PVAs identification groups. Overall, senior physicians had a significantly higher ability to correctly identify at least 3 PVAs than residents did (7 [53.8%] vs 14 [23.7%], $P = .044$). Moreover, higher physician:bed and resident:bed ratios were associated with better PVAs identification (1.42 [1.18–1.50] vs 1.25 [0.63–1.42] and 0.88 [0.64–0.90] vs 0.66 [0.40–0.88], respectively). Prior training in mechanical ventilation was not significantly associated with the ability to correctly identify at least 3 PVAs (17 [34%] vs 4 [18.2%], respectively, $P = .21$).

Discussion

The present study aimed to assess the ability of CCPs to identify different types of PVAs using visual waveform analysis and examine factors related to this ability. The results provided valuable insights into the proficiency of CCPs in visual waveform analysis and highlighted important considerations for optimizing PVA in critical care settings.

Whereas the overall accuracy rate demonstrated the poor ability of CCPs to recognize PVA patterns, it is noteworthy that accuracy rates differed among specific types of asynchrony. This finding suggests that certain types of PVAs may pose greater challenges in identification compared to others. Ineffective efforts and premature cycling were more challenging to discern accurately than double-triggering or auto-triggering. Further research could explore specific factors contributing to these differences and identify potential strategies for improving accuracy rates across all types of PVAs.

Factors related to identification ability among CCPs were also examined in this study. Although experience level was found to be a potential influencing factor on identification accuracy, no significant associations were observed between

Table 2. Factors Associated With Proper Identification of Patient-Ventilator Asynchronies Among Critical Care Physicians

	PVAs Proper Identification Group (n = 21)	PVAs Improper Identification Group (n = 51)	P
Scientific grade			.044
Resident	14 (23.7)	45 (76.3)	
Senior	7 (53.8)	6 (46.2)	
Prior training in mechanical ventilation			.20
Yes	17 (34)	33 (66)	
No	4 (18.2)	18 (81.8)	
Date of mechanical ventilation course			.60
< 1 y	9 (31)	20 (69)	
> 1 y	8 (38.1)	13 (61.9)	
Prior exposure to waveforms analysis			.08
Routine assessment	18 (36)	32 (64)	
Occasional assessment	3 (13.6)	19 (86.4)	
MICU patients admission/y			.17
< 500 patients	21 (31.8)	45 (68.2)	
≥ 500 patients	0	6 (100)	
MICU ventilated patients/y			.17
< 200 patients	18 (33.3)	36 (66.7)	
≥ 200 patients	3 (16.7)	15 (83.3)	
MICU bed capacity	10 [8–10]	12 [8–20]	.14
Physician:bed ratio	1.42 [1.18–1.50]	1.25 [0.63–1.42]	.045
Resident:bed ratio	0.88 [0.64–0.90]	0.66 [0.40–0.88]	.01
Senior:bed ratio	0.60 [0.50–0.62]	0.58 [0.30–0.60]	.43
Senior:resident ratio	0.66 [0.66–0.99]	0.87 [0.66–1.00]	.20

Data are presented as n (%) or median [interquartile range].

PVA = patient-ventilator asynchrony

MICU = medical ICU

prior training in mechanical ventilation, prior exposure to waveform analysis, and the ability to identify PVA. This suggests that experience plays a role in improving identification skills. However, it is important to note that even CCPs with less experience or with less educational backgrounds demonstrated slight accuracy, indicating the potential for skill development through training and education.

Strengths and Limitations

The present nationwide survey performed in a low middle-income country brings another demonstration of the limited knowledge regarding PVA identification among CCPs, especially juniors, even with prior training in mechanical ventilation and exposure to waveform analysis, but several limitations could be addressed. First, as mentioned, 2 blinded field experts labeled the studied waveforms. However, an alternative and intriguing approach that could have been explored is the “contrast with the norm” or “norm-referenced assessment” method, applied in

the pedagogy of medical education.²¹ This approach aims to assess students’ proficiency in comparison to an established norm or standard, defined by the performance of experts or a representative group. Unfortunately, expanding the group of experts could drastically reduce the pool of seniors included in the survey. Second, the design of the present study intentionally didn’t include a training session before the waveform analysis task to allow a real description of the state of knowledge on their prior training in mechanical ventilation and exposure to waveform analysis. Third, regarding the sample size, as the study addressed the whole population of CCPs, efforts have been made to maximize the response rate by sending multiple reminders to respondents. The final sample size didn’t allow us to perform multivariate analysis to identify factors independently associated with the ability to identify the PVAs. Fourth, our survey contained multiple-choice questions, which may have influenced the results since the CCPs could have randomly chosen the correct answer.

Also, whereas this study provides valuable insight into the ability of CCPs to detect PVA, the inclusion of multiple examples of each type of PVA, along with a normal waveform, would have been beneficial for identifying and removing any ambiguous waveforms that could have influenced their accurate identification. Nonetheless, we recognized that such an approach might have potentially compromised the overall response rate. Striking a balance between enhancing waveform identification and maintaining an acceptable response rate presented a challenging trade-off during the study design. Another limitation is that being used to a certain type of ventilator might influence the ability to correctly analyze waveforms on unfamiliar ones. Nevertheless, due to the extreme difficulty of ventilator acquisitions in Tunisia, it is near impossible to be able to standardize ventilators in the ICUs; and as a consequence, there are many different brands and models of ventilators in each ICU, with some dedicated to noninvasive ventilation and other for invasive ventilation only. Furthermore, critical care residents switch from one ICU to another every 6 to 12 months, thus experiencing many different types of ventilators. Additionally, one could argue that the ability to recognize certain PVAs could be enhanced by providing a video clip of multiple respiratory cycles; but by using simple screenshots including several cycles, we aimed to simplify the survey in order to increase the response rate. Finally, whereas the study assessing physicians in Tunisia could raise concern about its generalizability, it is essential to note that the recognition of asynchronies is a global concern.

Results Interpretation

Previous studies have reported a low rate of correct identification of PVAs, with the rate differing according to the

studied population. Ramirez et al¹⁸ reported that 21.3% of the respondents correctly recognized all types of asynchrony. Alqahtani et al²² found that 10.2% of physicians, nurses, and respiratory therapists recognized all types of asynchrony.

In the present study, the most frequently correctly identified PVA type was double-triggering (59.7%), followed by auto-triggering (50%). This is consistent with reports by Ramirez et al¹⁸ and Alqahtani et al²² indicating that double-triggering was the most frequently identified asynchrony (58.2% and 48.7%, respectively), followed by auto-triggering (49.4% and 32.7%, respectively). Ineffective triggering ranked third; however, a previous study reported that it was the most frequent asynchrony.¹⁸

The present study exhibited that certain patient-ventilator interactions are prone to being misidentified as specific types of asynchrony. The observed misidentification patterns imply that certain patient-ventilator interactions may share similar waveform characteristics, contributing to confusion among health care professionals during their interpretation. It is imperative to gain a deep understanding of these misidentification patterns, as it plays a pivotal role in enhancing the accuracy of waveform analysis and avoiding potential errors in patient care. This issue underscores the need for a more comprehensive approach to the identification of PVAs. This approach should begin with precise ventilator mode identification and extend to a step-by-step analysis of the various components of the respiratory cycle while considering the clinical presentation. Emphasizing a holistic and systematic assessment of PVAs can aid clinicians in making accurate identifications, leading to better patient outcomes and improved management of mechanical ventilation in critical care settings.

Notably, we observed that senior physicians had a better ability to correctly identify PVAs than residents did. This finding is consistent with the report by Colombo et al²³ in which expert physicians had a higher sensitivity in detecting PVAs than residents (28% vs 16%, respectively, $P = .03$). Additionally, prior training in mechanical ventilation, which was defined as having attended a workshop on mechanical ventilation involving a session on waveform analysis, was not associated with the ability of CCPs to correctly identify PVAs. This is inconsistent with previous reports.^{18,22,24} This finding could be attributed to study differences in variables such as the number of hours dedicated to training CCPs on identifying PVAs and the inclusion of practical sessions in training. Moreover, these programs may have lacked refresher sessions to sustain the ability of the CCPs to correctly identify PVAs. An international survey conducted in 20 countries demonstrated that attending courses for > 100 h facilitated proper recognition of PVAs.²⁵ Moreover, Chacón et al²⁴ reported that specific training on waveforms for 20 d improved the ability of nurses to recognize PVAs. Our findings suggest that the currently applied training method on

waveform analysis cannot sufficiently improve the ability of CCPs to correctly identify PVAs; therefore, more formal training should be initiated. Consistent with the findings of the present study, it has been reported that the number of beds could influence the ability of health care professionals to identify PVAs.²⁵

Clinical Relevance

The present study findings have important clinical and research implications. First, prior training in mechanical ventilation seems to be insufficient in allowing the correct identification of PVAs. Since PVAs influence patient outcomes, visual ventilator waveform analysis for mechanically ventilated patients by CCPs should be considered as important as invasive monitoring of arterial blood pressure or pulse contour continuous cardiac output in hemodynamically unstable patients,²⁶ invasive monitoring of intracranial pressure in patients with severe brain injury,²⁷ or monitoring of burst suppression in comatose or deeply sedated patients.²⁸ Therefore, CCPs should be trained in identifying PVAs in their daily clinical practice through different methods, including specialized bedside visits. Moreover, CCPs should receive frequent teaching and hands-on sessions on ventilator graphics rather than theoretical sessions on waveform analysis in a mechanical ventilation course. Comprehensive diagnostic approaches of visual waveform analysis could improve detection and identification in PVAs.²⁹⁻³² Such initiatives can enhance staff knowledge and ultimately improve patient outcomes. Additionally, future studies are warranted to investigate the effectiveness of this training for CCPs. The improved ability of CCPs to identify PVAs may inform the establishment of a protocol for identifying and managing PVAs that could reduce excessive sedation and improve patient outcomes, as previously reported.³³

Future Directions

Based on the study results, potential avenues for future research may include studies focused on developing standardized training programs, evaluating the impact of educational interventions on identification accuracy, or investigating the correlation between identification accuracy and patient outcomes. In addition to educational programs, it is crucial to explore and invest in other levels of assistance for CCPs. By expanding the support framework, we can provide comprehensive resources and tools that cater to their specific needs and challenges. First, implementing software within the ventilator to detect and offset the most frequent PVAs.^{12,34} Casagrande et al³⁵ proposed the pressure-flow breath representation to ease PVA identification. Second, encouraging the monitoring of electrical activity of the diaphragm (EA_{di}) or of the esophageal or transdiaphragmatic pressure signal in selected patients.³⁶⁻⁴⁰

Sinderby et al⁴¹ proposed an automated and standardized method (NeuroSync Index), based on the assessment and monitoring of the EA_{di}, to quantify asynchronies. Third, building noninvasive and costless machine learning and artificial intelligence–derived algorithms to assist CCPs in detecting and identifying PVAs. Silva et al⁴² demonstrated that displaying an artificial intelligence–generated work of breath curves next to pressure, flow, and volume curves could help identify PVAs. Su et al⁴³ proposed a remote network platform to help detect and identify PVAs.

Conclusions

Our results revealed a notable deficiency in the ability of CCPs to accurately identify PVAs. Senior physicians exhibited superior skills in correctly identifying PVAs compared to resident physicians. Surprisingly, our study did not find any correlation between prior training in mechanical ventilation and the capability to accurately identify PVAs. This suggests that the educational methods and training programs for identifying PVAs using ventilator graphics are insufficient in quality. The outcomes of this study will enhance our comprehension of CCPs' proficiency in analyzing waveform data and offer valuable insights for improving their competence in identifying and effectively managing PVAs.

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